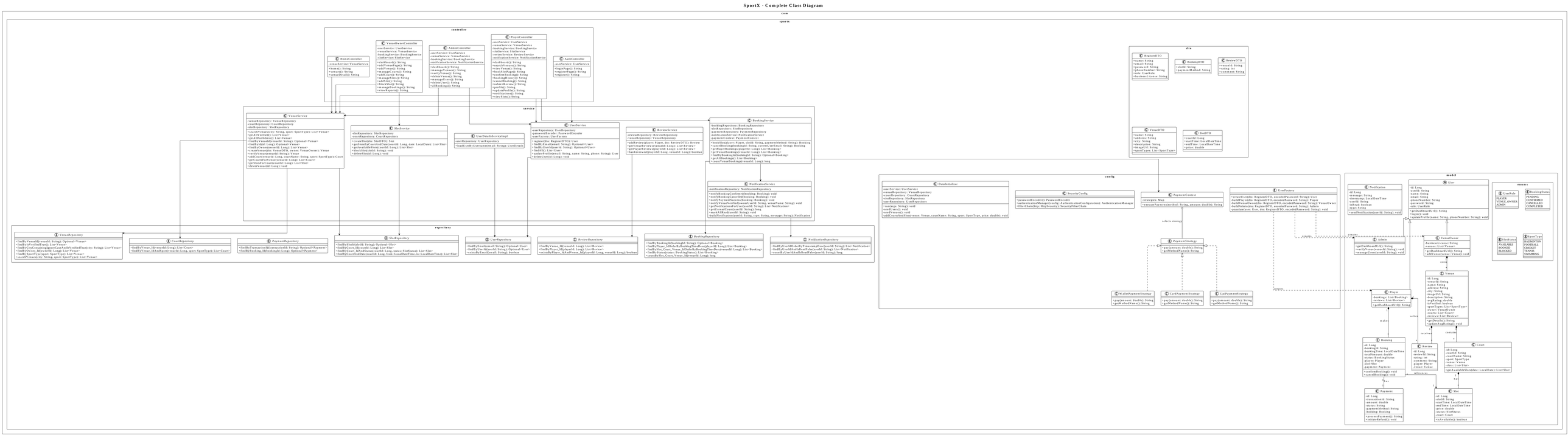

SportX - Complete Class Diagram

com
sportx
controller
HomeController
-venueService: VenueService
+home(): String
+venues(): String
+venueDetail(): String
VenueOwnerController
-userService: UserService
-venueService: VenueService
-bookingService: BookingService
-slotService: SlotService
+dashboard(): String
+addVenuePage(): String
+addVenue(): String
+manageCourts(): String
+addCourt(): String
+manageSlots(): String
+addSlot(): String
+blockSlot(): String
+manageBookings(): String
+viewReports(): String
AdminController
-userService: UserService
-venueService: VenueService
-bookingService: BookingService
-notificationService: NotificationService
+dashboard(): String
+manageVenues(): String
+verifyVenue(): String
+deleteVenue(): String
+manageUsers(): String
+deleteUser(): String
+allBookings(): String
PlayerController
-userService: UserService
-venueService: VenueService
-bookingService: BookingService
-slotService: SlotService
-reviewService: ReviewService
-notificationService: NotificationService
+dashboard(): String
+searchVenues(): String
+viewVenue(): String
+bookSlotPage(): String
+confirmBooking(): String
+bookingHistory(): String
+cancelBooking(): String
+submitReview(): String
+profile(): String
+updateProfile(): String
+notifications(): String
+viewSlots(): String
AuthController
-userService: UserService
+loginPage(): String
+registerPage(): String
+register(): String
service
VenueService
-venueRepository: VenueRepository
-courtRepository: CourtRepository
-slotRepository: SlotRepository
+searchVenues(city: String, sport: SportType): List<Venue>
+getAllVerified(): List<Venue>
+getAllForAdmin(): List<Venue>
+findByVenueId(venueId: String): Optional<Venue>
+findById(id: Long): Optional<Venue>
+findByOwner(ownerId: Long): List<Venue>
+createVenue(dto: VenueDTO, owner: VenueOwner): Venue
+verifyVenue(venueId: String): Venue
+addCourt(venueId: Long, courtName: String, sport: SportType): Court
+getCourtsForVenue(venueId: Long): List<Court>
+getSlotsForCourt(courtId: Long): List<Slot>
+deleteVenue(id: Long): void
SlotService
-slotRepository: SlotRepository
-courtRepository: CourtRepository
+createSlot(dto: SlotDTO): Slot
+getSlotsByCourtAndDate(courtId: Long, date: LocalDate): List<Slot>
+getAvailableSlots(courtId: Long): List<Slot>
+blockSlot(slotId: String): void
+deleteSlot(id: Long): void
UserDetailsServiceImpl
-userRepository: UserRepository
+loadUserByUsername(email: String): UserDetails
UserService
-userRepository: UserRepository
-passwordEncoder: PasswordEncoder
-userFactory: UserFactory
+register(dto: RegisterDTO): User
+findByEmail(email: String): Optional<User>
+findByUserId(userId: String): Optional<User>
+findAll(): List<User>
+updateProfile(email: String, name: String, phone: String): User
+deleteUser(id: Long): void
ReviewService
-reviewRepository: ReviewRepository
-venueRepository: VenueRepository
+addReview(player: Player, dto: ReviewDTO): Review
+getVenueReviews(venueId: Long): List<Review>
+getPlayerReviews(playerId: Long): List<Review>
+hasReviewed(playerId: Long, venueId: Long): boolean
BookingService
-bookingRepository: BookingRepository
-slotRepository: SlotRepository
-paymentRepository: PaymentRepository
-notificationService: NotificationService
-paymentContext: PaymentContext
+bookSlot(player: Player, slotId: String, paymentMethod: String): Booking
+cancelBooking(bookingId: String, currentUserEmail: String): Booking
+getPlayerBookings(playerId: Long): List<Booking>
+getVenueBookings(venueId: Long): List<Booking>
+findByBookingId(bookingId: String): Optional<Booking>
+getAllBookings(): List<Booking>
+countVenueBookings(venueId: Long): long
NotificationService
-notificationRepository: NotificationRepository
+notifyBookingConfirmed(booking: Booking): void
+notifyBookingCancelled(booking: Booking): void
+notifyPaymentSuccess(booking: Booking): void
+notifyVenueVerified(ownerUserId: String, venueName: String): void
+getNotificationsForUser(userId: String): List<Notification>
+getUnreadCount(userId: String): long
+markAllRead(userId: String): void
-buildNotification(userId: String, type: String, message: String): Notification
repository
VenueRepository
+findByVenueId(venueId: String): Optional<Venue>
+findByIsVerifiedTrue(): List<Venue>
+findByCityContainingIgnoreCaseAndIsVerifiedTrue(city: String): List<Venue>
+findByOwner_Id(ownerId: Long): List<Venue>
+findBySportType(sport: SportType): List<Venue>
+searchVenues(city: String, sport: SportType): List<Venue>
CourtRepository
+findByVenue_Id(venueId: Long): List<Court>
+findByVenue_IdAndSport(venueId: Long, sport: SportType): List<Court>
PaymentRepository
+findByTransactionId(transactionId: String): Optional<Payment>
+findByBooking_Id(bookingId: Long): Optional<Payment>
SlotRepository
+findBySlotId(slotId: String): Optional<Slot>
+findByCourt_Id(courtId: Long): List<Slot>
+findByCourt_IdAndStatus(courtId: Long, status: SlotStatus): List<Slot>
+findByCourtAndDate(courtId: Long, from: LocalDateTime, to: LocalDateTime): List<Slot>
UserRepository
+findByEmail(email: String): Optional<User>
+findByUserId(userId: String): Optional<User>
+existsByEmail(email: String): boolean
ReviewRepository
+findByVenue_Id(venueId: Long): List<Review>
+findByPlayer_Id(playerId: Long): List<Review>
+existsByPlayer_IdAndVenue_Id(playerId: Long, venueId: Long): boolean
BookingRepository
+findByBookingId(bookingId: String): Optional<Booking>
+findByPlayer_IdOrderByBookingTimeDesc(playerId: Long): List<Booking>
+findBySlot_Court_Venue_IdOrderByBookingTimeDesc(venueId: Long): List<Booking>
+findByStatus(status: BookingStatus): List<Booking>
+countBySlot_Court_Venue_Id(venueId: Long): long
NotificationRepository
+findByUserIdOrderByTimestampDesc(userId: String): List<Notification>
+findByUserIdAndIsReadFalse(userId: String): List<Notification>
+countByUserIdAndIsReadFalse(userId: String): long
dto
RegisterDTO
+name: String
+email: String
+password: String
+phoneNumber: String
+role: UserRole
+businessLicense: String
BookingDTO
+slotId: String
+paymentMethod: String
ReviewDTO
+venueId: String
+rating: int
+comment: String
VenueDTO
+name: String
+address: String
+city: String
+description: String
+imageUrl: String
+sportTypes: List<SportType>
SlotDTO
+courtId: Long
+startTime: LocalDateTime
+endTime: LocalDateTime
+price: double
config
DataInitializer
-userService: UserService
-venueRepository: VenueRepository
-courtRepository: CourtRepository
-slotRepository: SlotRepository
-userRepository: UserRepository
+run(args: String): void
-seedUsers(): void
-seedVenues(): void
-addCourtsAndSlots(venue: Venue, courtName: String, sport: SportType, price: double): void
SecurityConfig
+passwordEncoder(): PasswordEncoder
+authenticationManager(config: AuthenticationConfiguration): AuthenticationManager
+filterChain(http: HttpSecurity): SecurityFilterChain
PaymentContext
-strategies: Map
+executePayment(method: String, amount: double): String
selects strategy
PaymentStrategy
+pay(amount: double): String
+getMethodName(): String
WalletPaymentStrategy
+pay(amount: double): String
+getMethodName(): String
CardPaymentStrategy
+pay(amount: double): String
+getMethodName(): String
UpiPaymentStrategy
+pay(amount: double): String
+getMethodName(): String
UserFactory
+createUser(dto: RegisterDTO, encodedPassword: String): User
-buildPlayer(dto: RegisterDTO, encodedPassword: String): Player
-buildVenueOwner(dto: RegisterDTO, encodedPassword: String): VenueOwner
-buildAdmin(dto: RegisterDTO, encodedPassword: String): Admin
-populate(user: User, dto: RegisterDTO, encodedPassword: String): void
creates
creates
creates
model
Notification
-id: Long
-message: String
-timestamp: LocalDateTime
-userId: String
-isRead: boolean
-type: String
+sendNotification(userId: String): void
User
-id: Long
-userId: String
-name: String
-email: String
-phoneNumber: String
-password: String
-role: UserRole
+getDashboardUrl(): String
+login(): void
+updateProfile(name: String, phoneNumber: String): void
Admin
+getDashboardUrl(): String
+verifyVenue(venueId: String): void
+manageUsers(userId: String): void
VenueOwner
-businessLicense: String
-venues: List<Venue>
+getDashboardUrl(): String
+addVenue(venue: Venue): void
1
owns
*
Venue
-id: Long
-venueId: String
-name: String
-address: String
-city: String
-imageUrl: String
-description: String
-avgRating: double
-isVerified: boolean
-sportTypes: List<SportType>
-owner: VenueOwner
-courts: List<Court>
-reviews: List<Review>
+getDetails(): String
+updateAvgRating(): void
Player
-bookings: List<Booking>
-reviews: List<Review>
+getDashboardUrl(): String
1
makes
*
1
writes
*
1
receives
*
1
contains
*
Booking
-id: Long
-bookingId: String
-bookingTime: LocalDateTime
-totalAmount: double
-status: BookingStatus
-player: Player
-slot: Slot
-payment: Payment
+confirmBooking(): void
+cancelBooking(): void
Review
-id: Long
-reviewId: String
-rating: int
-comment: String
-player: Player
-venue: Venue
Court
-id: Long
-courtId: String
-courtName: String
-sport: SportType
-venue: Venue
-slots: List<Slot>
+getAvailableSlots(date: LocalDate): List<Slot>
*
references
1
1
has
1
1
has
*
Payment
-id: Long
-transactionId: String
-amount: double
-status: String
-paymentMethod: String
-booking: Booking
+processPayment(): String
+initiateRefund(): void
Slot
-id: Long
-slotId: String
-startTime: LocalDateTime
-endTime: LocalDateTime
-price: double
-status: SlotStatus
-court: Court
+isAvailable(): boolean
enums
UserRole
PLAYER
VENUE_OWNER
ADMIN
BookingStatus
PENDING
CONFIRMED
CANCELLED
COMPLETED
SlotStatus
AVAILABLE
BOOKED
BLOCKED
SportType
BADMINTON
FOOTBALL
CRICKET
TENNIS
SWIMMING